

AI Resume Analyzer Project Report

1. Introduction

Getting a job these days is really tough and resume is the first thing that matters. I noticed that many students don't know what to write in their resume or how to make it better — ye problem mujhe bhi thi. So I decided to build something that can analyze a resume and give quick feedback using machine learning.

2. Problem

Most students don't know what skills to add in their resume. Manually checking resumes also takes a lot of time. So I thought why not make something that does this automatically. Chhota project hai but the idea is real.

3. What I Wanted to Do

- read resume text and tell if it's strong or weak
 - detect skills from the resume
 - show a confidence score
 - and honestly — ml seekhna tha toh this was a good excuse 😊
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4. How I Built It

First I created a small dataset manually with some resume examples and labeled them as selected or rejected. Ye part thoda boring tha but had to do it.

Then I used TF-IDF to convert text into numbers because machine learning models can't read plain text directly. This concept took me some time to understand — watched a few YouTube videos and then it made sense.

For the model I used Naive Bayes classifier. It's simple and works well for text data. For the interface I used Streamlit because HTML CSS nahi aati and Streamlit is much faster to build with.

5. Tools I Used

python, pandas, sklearn, streamlit — nothing fancy.

6. Project Structure

Divided into 3 files:

- **train.py** — loads data, trains model, saves it
 - **utils.py** — prediction and skill detection logic
 - **app.py** — the interface part jo user dekhta hai
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7. What It Can Do

- tells if resume is strong or needs improvement
 - gives a confidence score
 - detects skills like python, sql, machine learning etc
 - simple ui, easy to use
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8. Results

It works. When I paste a resume it gives prediction and shows detected skills. Accuracy is not perfect because dataset is small but for a beginner project it's decent enough I think.

9. Problems I Faced

- TF-IDF samajh nahi aa raha tha at first, had to watch videos
 - creating the dataset was hard, didn't know how many examples are enough
 - file path errors were really frustrating — model not found kept coming
 - also took time to understand how pickle works
 - some Streamlit buttons were not working properly, bahut time laga fix karne mein
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10. What I Learned

- how machine learning models are trained practically
- what TF-IDF does and why it's needed
- how to save and load models using pickle

- how to build a basic app with Streamlit
 - and the biggest lesson — debugging mein hi adha time jaata hai 😊
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11. Conclusion

This was my first properly working machine learning project and I'm glad I completed it. It's not perfect but it works and I built it myself — stack overflow aur YouTube ki help se hi sahi 😊. I want to improve it more in the future.

12. Future Plans

- add pdf and docx upload feature
 - build a bigger dataset for better accuracy
 - add job role based suggestions
 - improve the ui, abhi bahut basic hai
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13. Thanks

This project was part of my AIML course. Thanks to my teachers for the guidance. Aur Google aur YouTube ka bhi shukriya — unke bina ye possible nahi hota 😊

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